

Dataset Analysis Approach

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address specific analytical questions, I first need to understand the structure and contents of the `sicor_municipal_contracts.csv` file. Below is a step-by-step breakdown of how I will approach this task:

Step 1: Load the Dataset

I will load the dataset using Python's `pandas` library to examine its structure, including column names, data types, and any missing values.

Step 2: Explore Key Characteristics

I will identify key characteristics of the dataset, such as:

- Number of rows and columns
- Column names and their descriptions (if available)
- Data types of each column
- Presence of missing or null values
- Summary statistics for numerical columns

Step 3: Address the Analytical Question

Once you provide a specific analytical question, I will:

1. Extract relevant data from the dataset.
2. Process the data to compute the required metrics or insights.
3. Provide a clear and concise answer to the question.
4. Include the Python code used for analysis.

Initial Analysis Code

Below is the Python code to load and explore the dataset:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values per Column:")
17 print(data.isnull().sum())
18
```



```
19 # Summary statistics for numerical columns
20 print("\nSummary Statistics for Numerical Columns:")
21 print(data.describe())
```

Expected Output

The above code will provide:

1. **Dataset Overview** : Information about the number of rows, columns, and data types.
2. **First Few Rows** : A preview of the dataset to understand its structure.
3. **Missing Values** : Identification of columns with missing data.
4. **Summary Statistics** : Insights into numerical columns (e.g., mean, min, max).

Next Steps

Please provide a specific analytical question you would like addressed using this dataset. For example:

- What is the total value of contracts awarded in each municipality?
- Which contractor received the highest total contract value?
- How many contracts were awarded in each year?

Once you provide the question, I will proceed with the detailed analysis and deliver the results along with the corresponding Python code.

What were the highest and lowest total contract values?



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To answer the question about the highest and lowest total contract values, I will follow these steps:

Step 1: Identify the Relevant Column

The dataset likely contains a column that represents the value of each contract (e.g., `contract_value` or similar). I will identify this column and ensure it is numeric for analysis.

Step 2: Compute the Highest and Lowest Values

Using the identified column:

- The **highest total contract value** will be the maximum value in the column.
- The **lowest total contract value** will be the minimum value in the column.

Step 3: Implement the Analysis in Python

Below is the Python code to compute the highest and lowest total contract values:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sior_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Identify the column representing contract values (assuming it's named 'contract_value')
8 # If the column has a different name, replace 'contract_value' with the correct name
9 if 'contract_value' in data.columns:
10     # Ensure the column is numeric
11     data['contract_value'] = pd.to_numeric(data['contract_value'], errors='coerce')
12
13     # Compute the highest and lowest total contract values
14     highest_value = data['contract_value'].max()
15     lowest_value = data['contract_value'].min()
16
17     # Display the results
18     print(f"Highest Total Contract Value: {highest_value}")
19     print(f"Lowest Total Contract Value: {lowest_value}")
20 else:
21     print("Error: The dataset does not contain a 'contract_value' column. Please verify the column name.")
```

Explanation of the Code

1. **Loading the Dataset** : The dataset is loaded using `pandas.read_csv` .
2. **Identifying the Column** : The code assumes the column name is `contract_value` . If the column name differs, you should replace it with the correct one.
3. **Ensuring Numeric Data** : The `pd.to_numeric` function ensures the column is treated as numeric, converting any non-numeric values to `NaN` .
4. **Computing Metrics** : The `max()` and `min()` functions are used to find the highest and lowest values, respectively.
5. **Output** : The results are printed in a clear format.

Expected Output

If the dataset contains a valid `contract_value` column, the output will look like this:

```
1 Highest Total Contract Value: <value>
2 Lowest Total Contract Value: <value>
```

If the column name is incorrect or missing, the code will prompt an error message.

Next Steps

Please confirm the name of the column that contains contract values. If it is not `contract_value` , provide the correct column name so I can adjust the code accordingly. Alternatively, if you have already verified the column name, let me know, and I will proceed with the analysis.

